

Quadrature Hybrid Integrated Dual-Circularly Polarized Array Antenna

Muhammad Asad Rahman, Eisuke Nishiyama, and Ichihiko Toyoda

Graduate School of Science and Engineering

Saga University

1 Honjo-machi, Saga-shi, Saga 840-8502, Japan

asad31@ceng.ec.saga-u.ac.jp, nishiyama@ceng.ec.saga-u.ac.jp, toyoda@cc.saga-u.ac.jp

Abstract—A new dual-circularly polarized array antenna integrated with a quadrature hybrid is presented in this paper. The antenna shows wideband performance due to the wideband characteristic of the quadrature hybrid. Two substrate layers and three metal layers are used to realize the proposed antenna. Both-sided MIC technology is used to distribute the whole feed network on the different substrate layers. Use of this technique leads to compact structure of the proposed array antenna. Two orthogonal modes with quadrature phase shift to excite circular polarization is achieved by using orthogonally fed patches and quadrature hybrid. Moreover, use of quadrature hybrid enables to change the circular polarization sense. The performances of the proposed antenna is experimentally validated and measured axial ratios are more than 9%.

Keywords—Array antenna, circular polarization, microstrip antenna, quadrature hybrid, wideband

I. INTRODUCTION

Circularly polarized (CP) antennas are widely used for many modern wireless communication applications such as satellite communication, radar, and GPS. In case of CP antennas, no direct alignment is needed between transmitting and receiving antennas that makes this kind of antennas attractive for modern communication applications.

CP can be excited by creating perturbation on the patch or feeding the patch by two orthogonal feed ports with quadrature phase shift. Many microstrip array antennas have been investigated for dual-CP operation such as an array antenna employing perturbed patch elements with interlaced sequential rotated feed [1], array with orthogonal linearly polarized patches and a 90° hybrid circuit [2], ring-slot array antenna integrated with de Ronde's coupler [3], and ring-slot array antenna with orthogonal feed circuit [4].

In this study, a novel orthogonally fed 1×2 dual-CP antenna is demonstrated using multi-layer structure. A quadrature hybrid is integrated to obtain a CP diversity function. Consequently, the array exhibits wideband performance due to the wideband characteristics of the quadrature hybrid. Slot lines on the mid layer are used to make a coupling between the quadrature hybrid of bottom layer and microstrip feed lines of the patches on the top layer. Both-sided MIC technology helps to make the structure compact by placing the quadrature hybrid just below the patches. Hence, the proposed antenna gives a compact structure compared to conventional orthogonally fed array antenna at the cost of multi-layer structure.

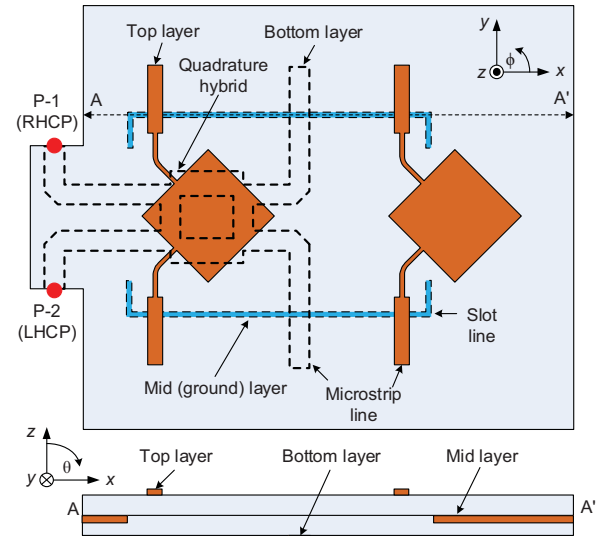
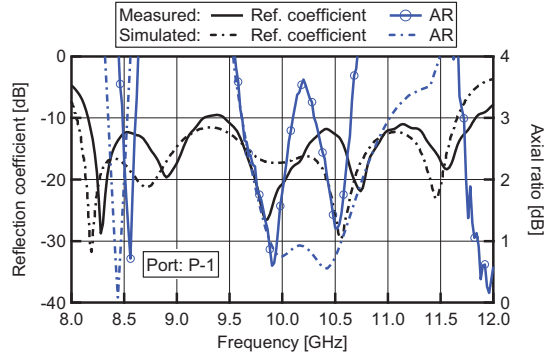


Fig. 1. Schematic layout and cross sectional view (AA') of the proposed antenna.

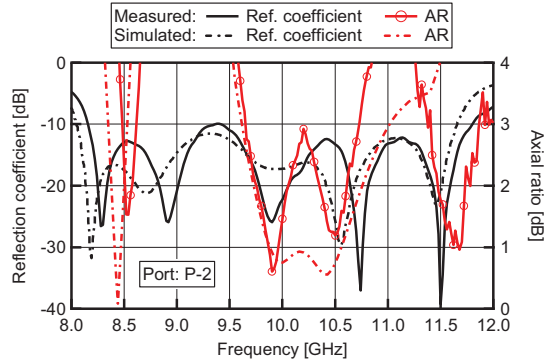
II. STRUCTURE

Fig. 1 shows the schematic structure and cross sectional view of the proposed array antenna. The antenna consists of three metal layers. Patch elements are on the top layer. Each patch element is excited by two 45° inclined feed lines which are orthogonal to each other. On the other hand, a quadrature hybrid is used to realize 90° phase difference between the feed lines of the patches. The quadrature hybrid is placed on the bottom layer of the substrate. The antenna structure becomes compact due to the placement of the quadrature hybrid just beneath the patches. The hybrid is designed for $50\text{-}\Omega$ microstrip lines to reduce the number of impedance transformers. The whole feed structure of the antenna is designed by applying both-sided MIC technology that helps to place two slot lines on the mid layer. These two slot lines couple the microstrip feed lines of the other two layers. The mid metal layer also acts as ground plane of the antenna.

In the proposed antenna, two orthogonal modes and quadrature phase shift are achieved by 45° inclined feed lines and quadrature hybrid, respectively to excite CP. Quadrature phase shift between two inclined feed lines can be altered by interchanging the feed ports, i.e. P-1 and P-2 that lead to the excitation of RHCP and LHCP, respectively.



(a) Port: P-1



(b) Port: P-2

Fig. 2. Reflection coefficient and axial ratio of the proposed antenna.

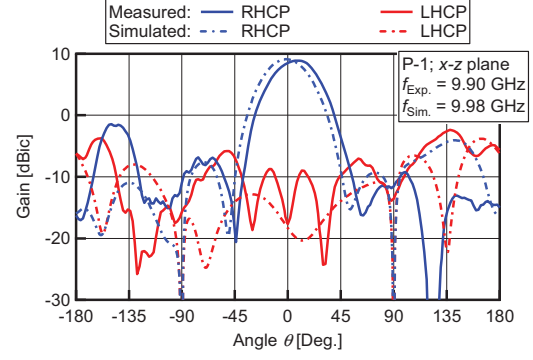
III. PERFORMANCES

The proposed antenna has been fabricated on a 0.8-mm thick Teflon glass fiber substrate with a dielectric permittivity of 2.15. Simulation has been carried out using Keysight Technologies' ADS considering infinite ground plane. Fig. 2(a) and (b) present the measured and simulated reflection coefficient and axial ratio (AR) of the proposed antenna. The measured 10-dB impedance bandwidths for P-1 and P-2 are around 34.89% and 36.47%, respectively. The measured 3-dB AR bandwidth for P-1 is more than 9%. On the other hand, a measured 3-dB AR bandwidth of 12.24% is achieved at P-2. Therefore, a wideband performance has been obtained. A good agreement is observed between measured and simulated reflection coefficients of P-1 and P-2. On the contrary, a slight disagreement between measured and simulated AR is noticed due to the formation of the airgap and the small shift between substrate layers.

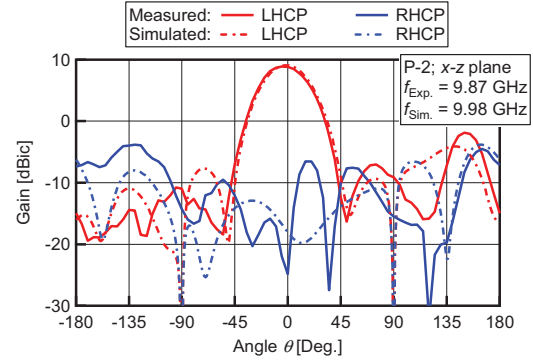
Fig. 3(a) and (b) show the measured and simulated radiation patterns of the antenna for P-1 and P-2, correspondingly. The patterns are plotted at a frequency where AR value is minimum. At P-1, the polarization sense is RHCP and measured maximum gain is 8.5 dBic. On the other hand, reverse CP sense can be excited at P-2 while a maximum gain of 8.86 dBic is obtained. Good radiation performances are observed with a cross polarization level better than 25 dB.

IV. CONCLUSION

This paper presents a multi-layer orthogonal feed microstrip array antenna integrated with a quadrature hybrid for dual-CP



(a) Port: P-1



(b) Port: P-2

Fig. 3. Radiation pattern of the proposed antenna.

operation. The antenna exhibits a wideband performance. Both-sided MIC technology helps to distribute the feed networks and quadrature hybrid in different layers. Therefore, the flexibility of placing 90° hybrid just underneath the patches makes the antenna more compact than the similar kind of conventional array antenna.

ACKNOWLEDGMENT

The authors would like to thank Dr. Takayuki Tanaka, Saga University for his fruitful discussions.

REFERENCES

- [1] Y. Shen, S.-G. Zhou, G.-L. Huang, and T.-H. Chio, "A compact dual circularly polarized microstrip patch array with interlaced sequentially rotated feed," *IEEE Trans. Antennas Propag.*, vol. 64, no. 11, pp. 4933-4936, Nov. 2016.
- [2] Y. Ushijima, E. Nishiyama, and M. Aikawa, "Single-layer integrated microstrip array antenna for dual circular polarisation," *IET Microw. Antennas Propag.*, vol. 6, no. 8, pp. 962-968, Jun. 2012.
- [3] S. Feng, E. Nishiyama, and M. Aikawa, "Broad-band circularly polarized ring-slot array antenna for simultaneous use of the orthogonal polarizations," *IEICE Trans. Electron.*, vol. E93-C, no. 7, pp. 1105-1110, Jul. 2010.
- [4] Y. Ushijima, S. Feng, E. Nishiyama, and M. Aikawa, "A novel circular polarization switchable slot-ring array antenna with orthogonal feed circuit," in *Proc. Asia-Pacific Microw. Conf.*, 2010, pp. 1569-1572.